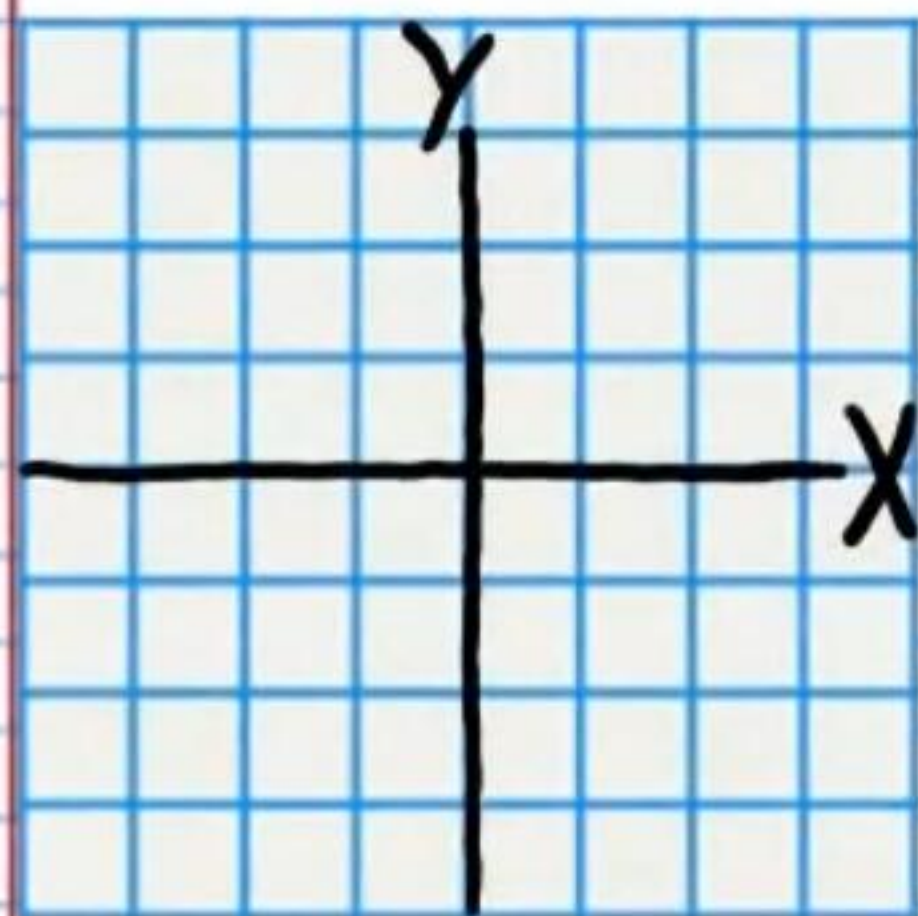


Graphing Polar Equations and Converting to Rectangular Form (Homework)

Graph the following equations. Write the coordinates of at least 4 points. Use a graphing calculator to draw the other parts of the plot.

① $r = \sqrt{95} \sin(2\theta)$

* Hint: We are not allowed to have a negative number under a square root symbol, so we must choose θ values between 0° & 90° or between 180° & 270° .



1st Point

2nd Point Try 195°



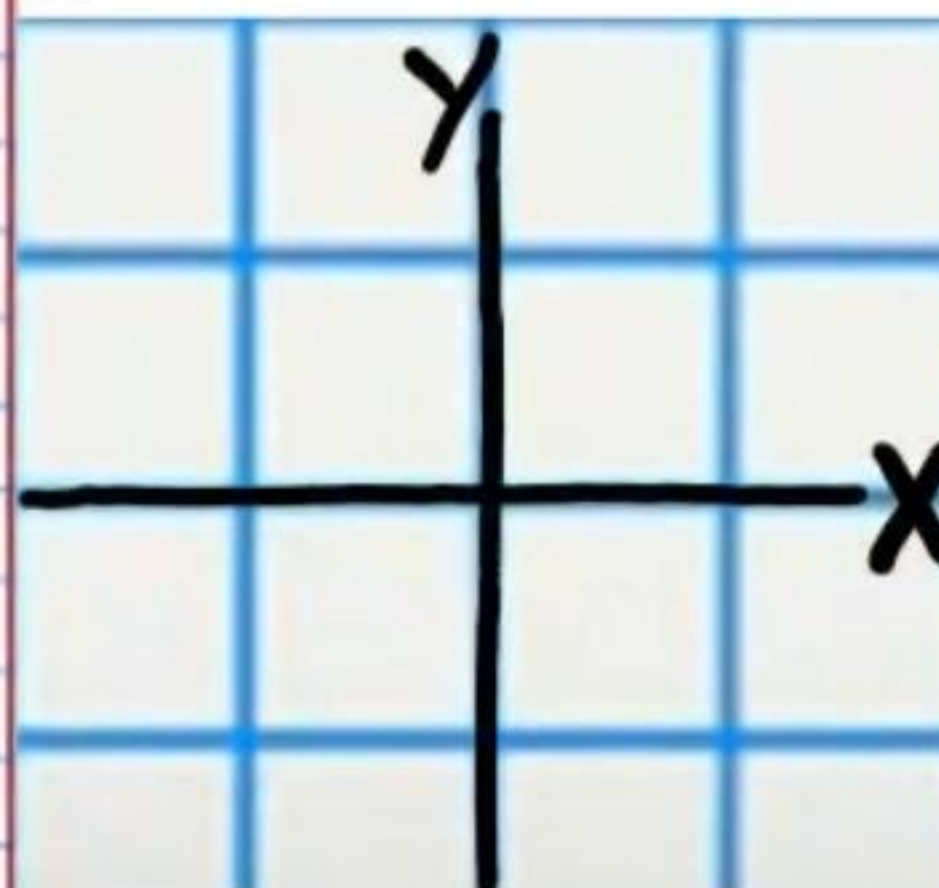
3rd Point

4th Point

② $r = 1 - \cos\theta$

1st Point

2nd Point



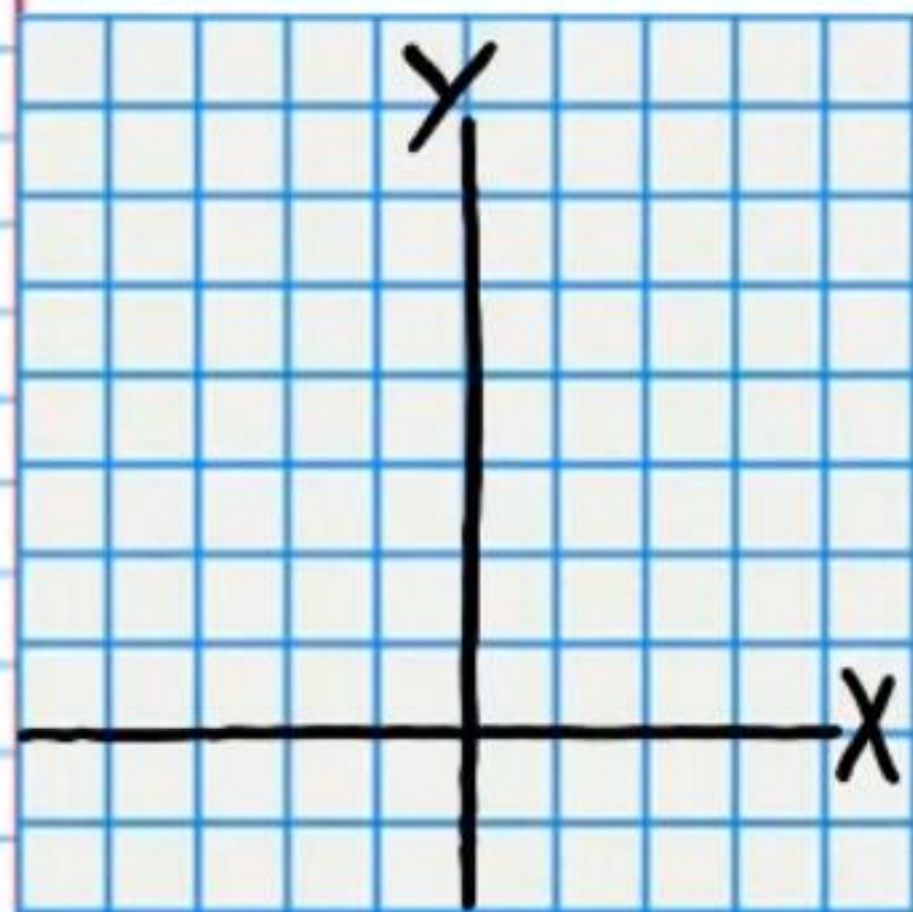
3rd Point

4th Point

③ $r = 6 \sin \theta$

1st Point

2nd Point



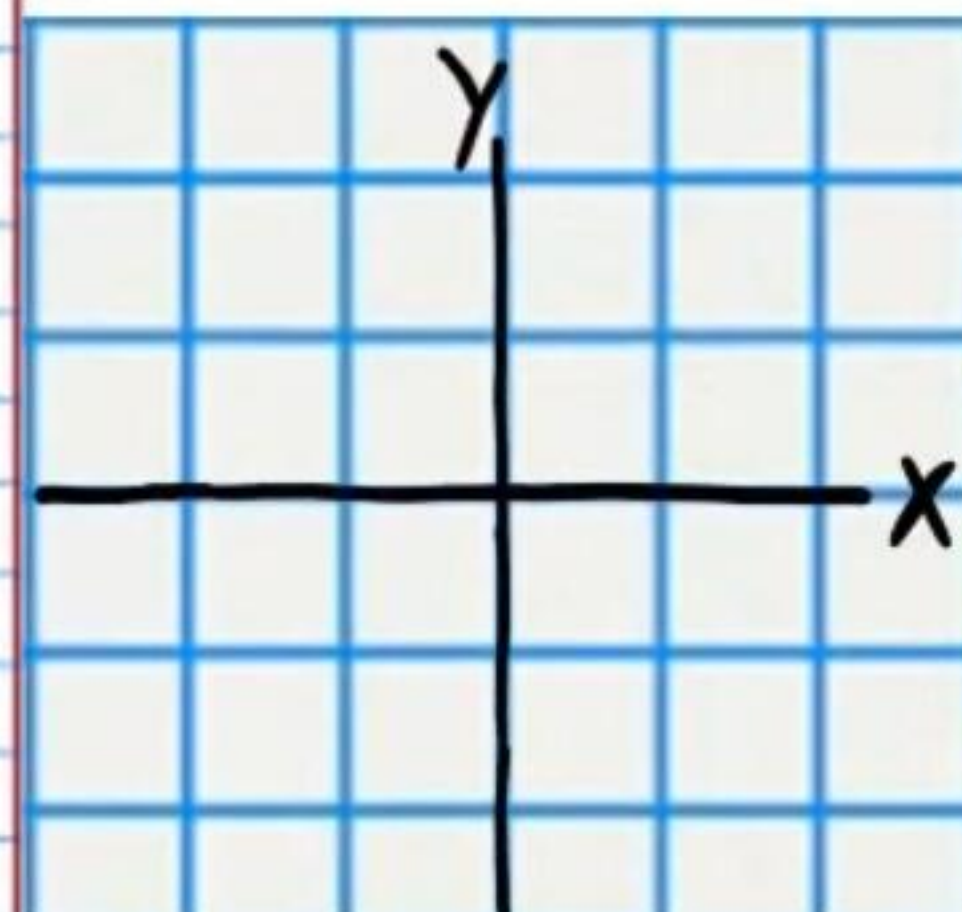
3rd Point

4th Point

④ $r = 2 \cos(2\theta)$

1st Point

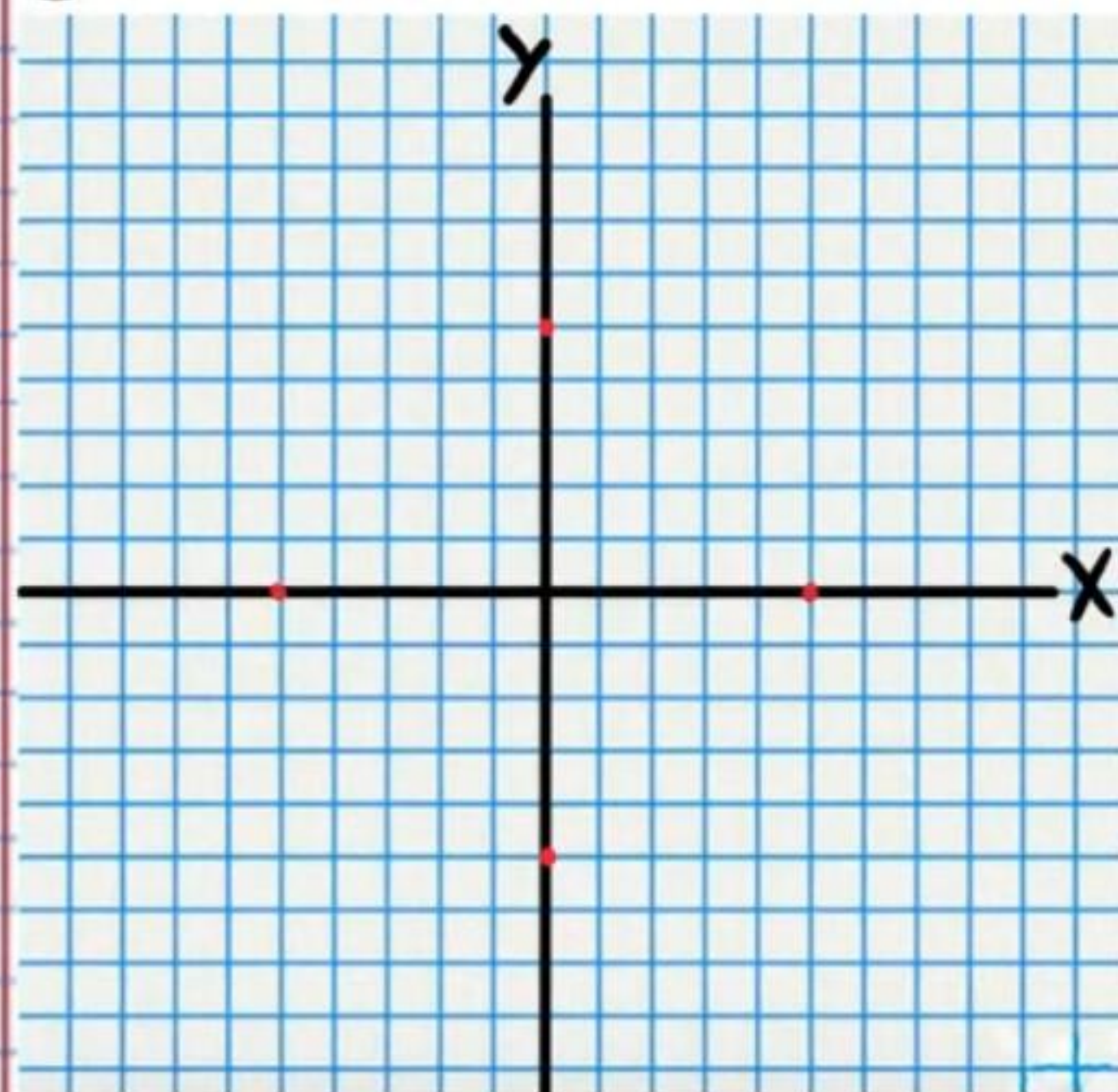
2nd Point



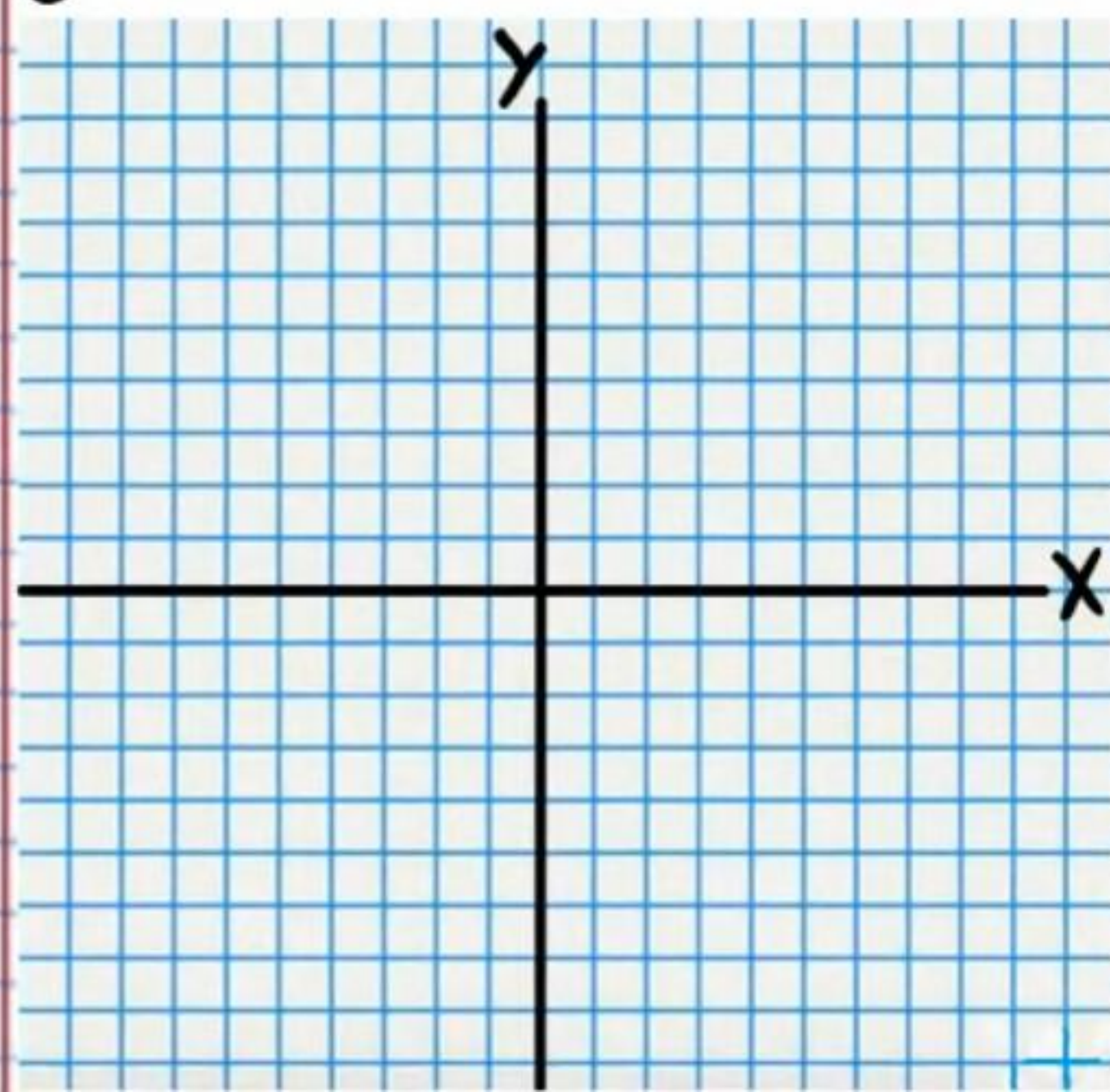
3rd Point

4th Point

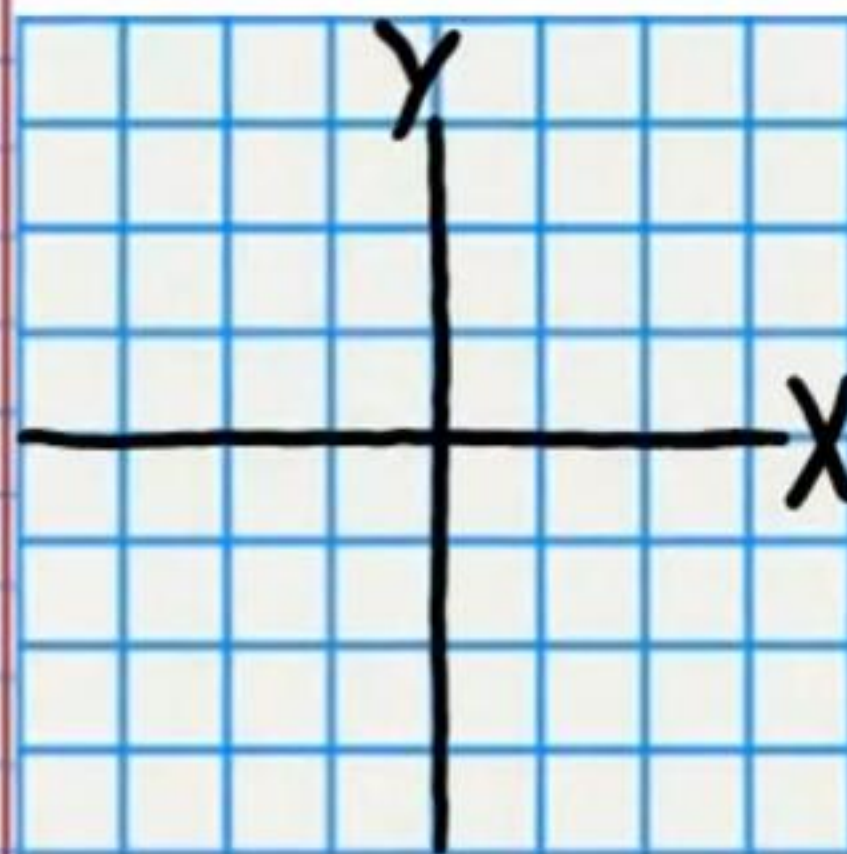
⑤ $r=5$



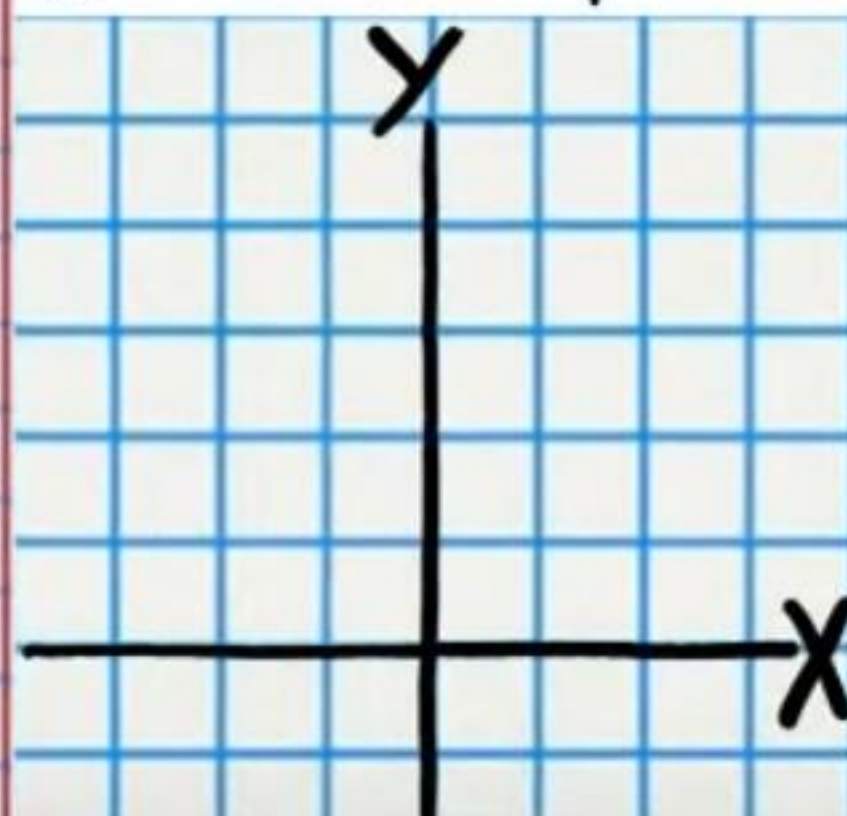
⑥ $r=8$



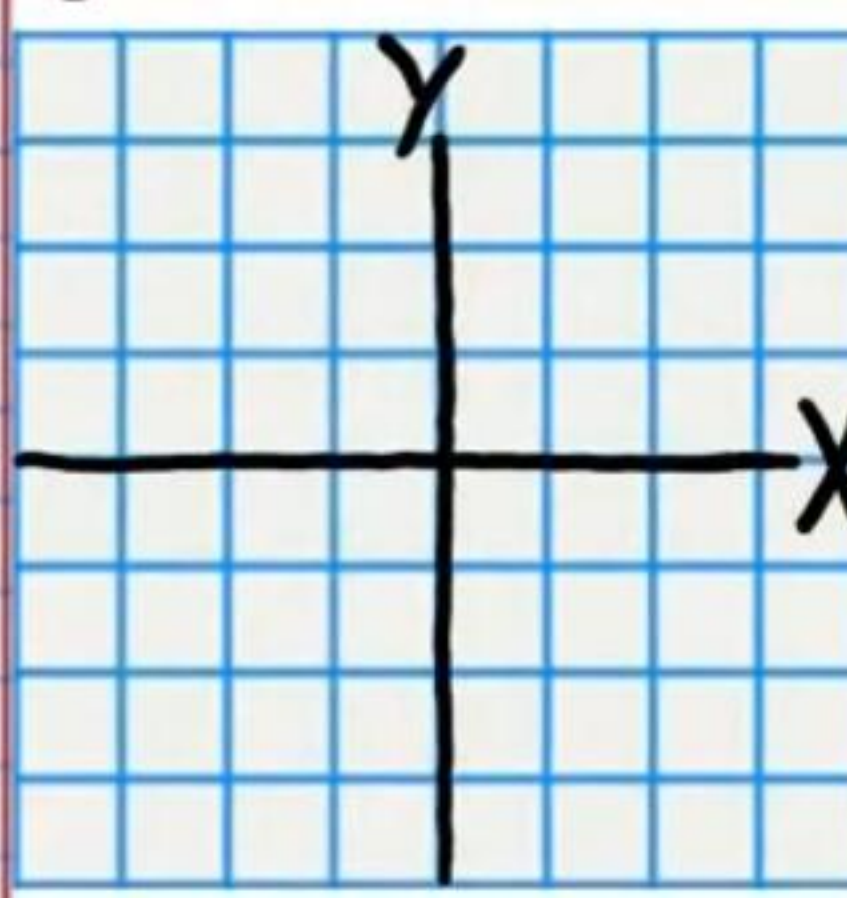
⑦ $r \sin \theta = 1$



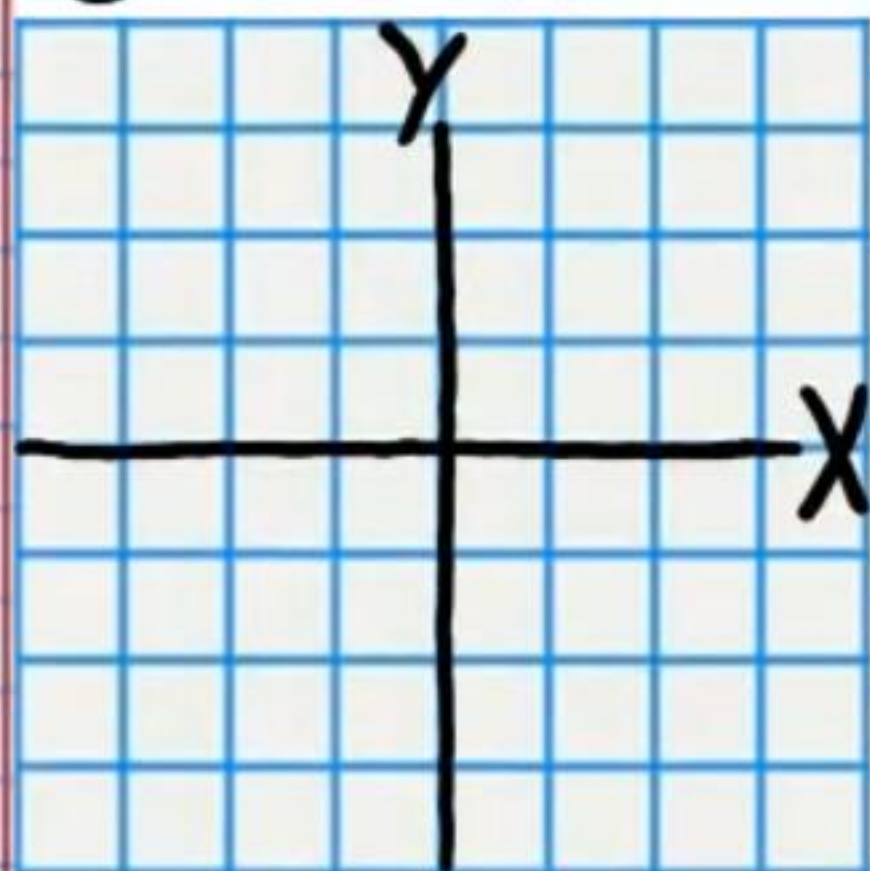
⑧ $\cos \theta = \frac{2}{r}$



⑨ $\theta = -150^\circ$



⑩ $\theta = 132^\circ$



Convert to rectangular form.

⑪ $r \sin \theta = \sqrt{r \cos \theta}$

⑫ $r^2 = 1 + \tan \theta$

⑬ $\cos \theta + \sin \theta = r$

⑭ $r = 4 + \sin \theta$

Convert to polar form.

⑮ $y = 100$

⑯ $4x + 2y = x^2 + y^2$

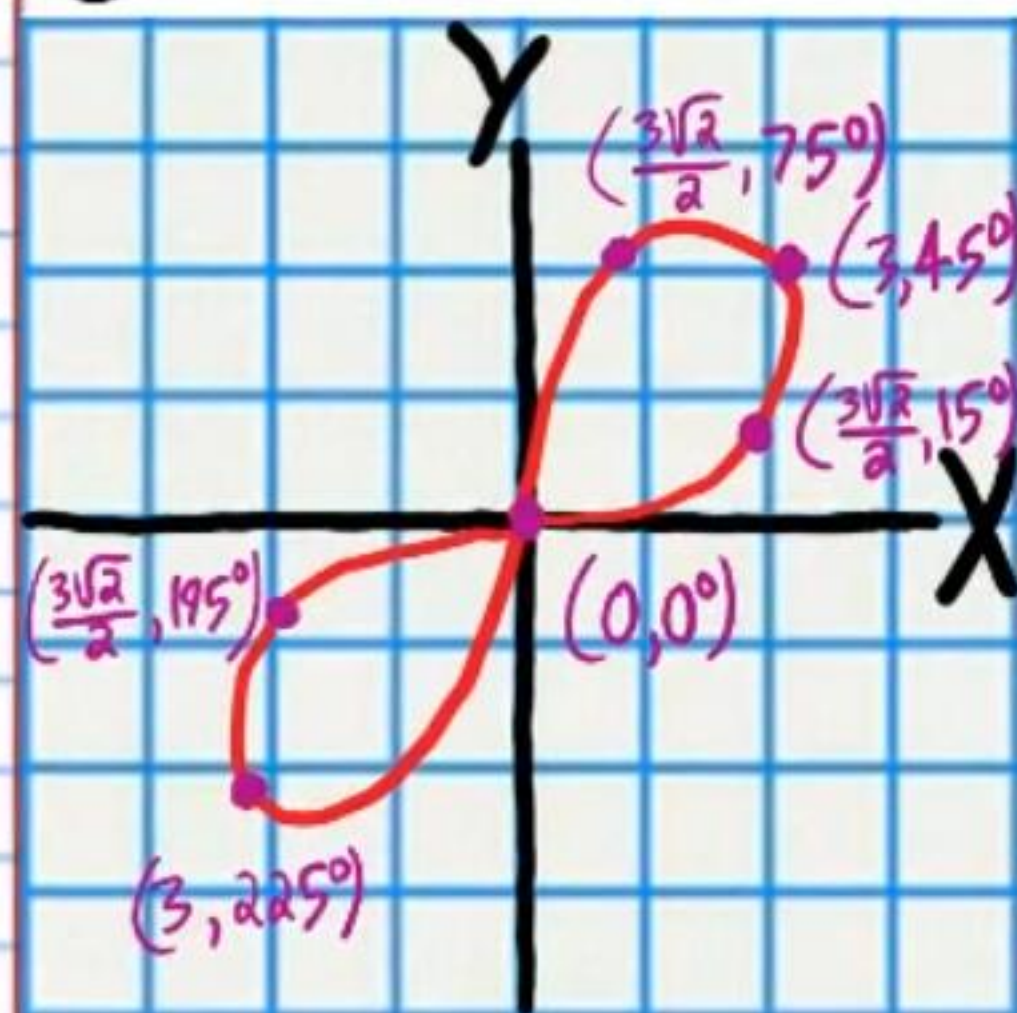
⑰ $y = x$

⑱ $1 = xy$

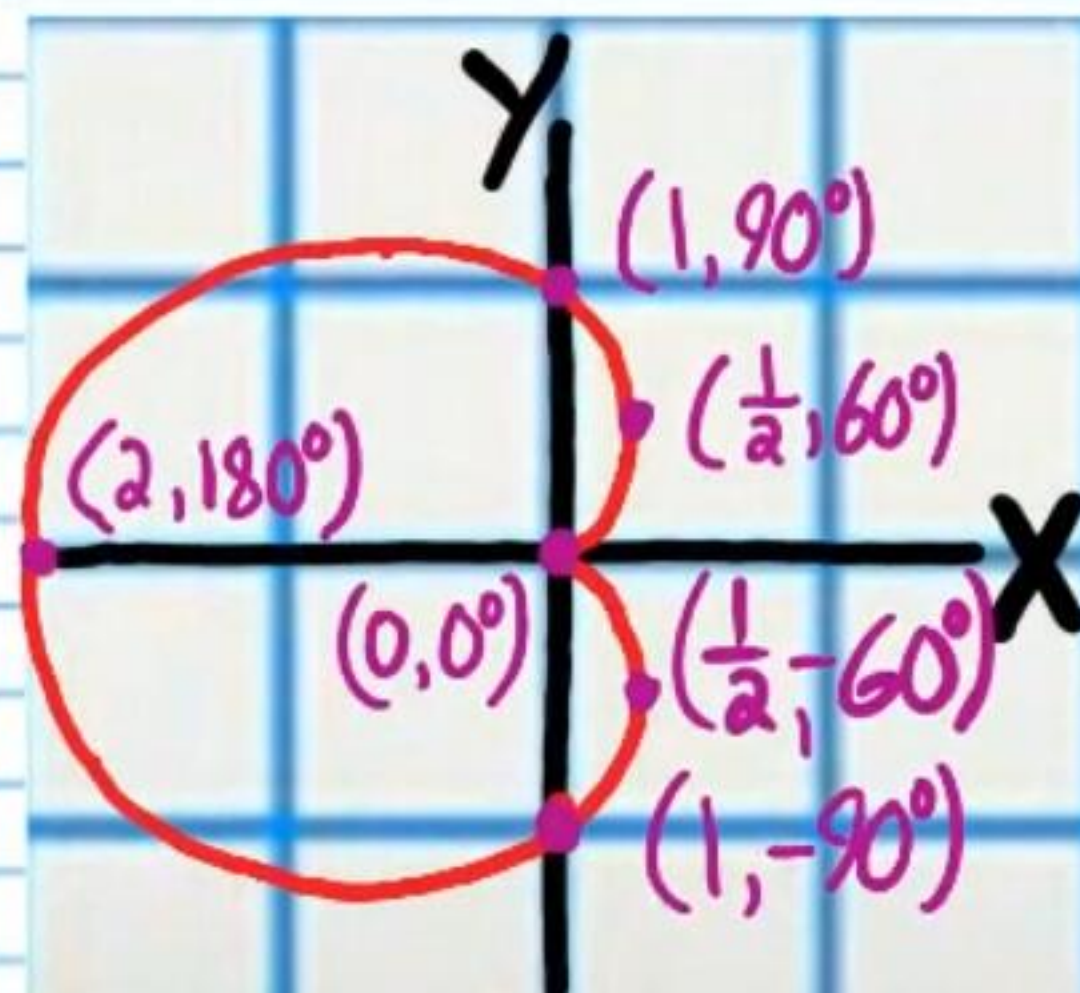
Homework Answers

* Many points are shown, but only four are necessary.

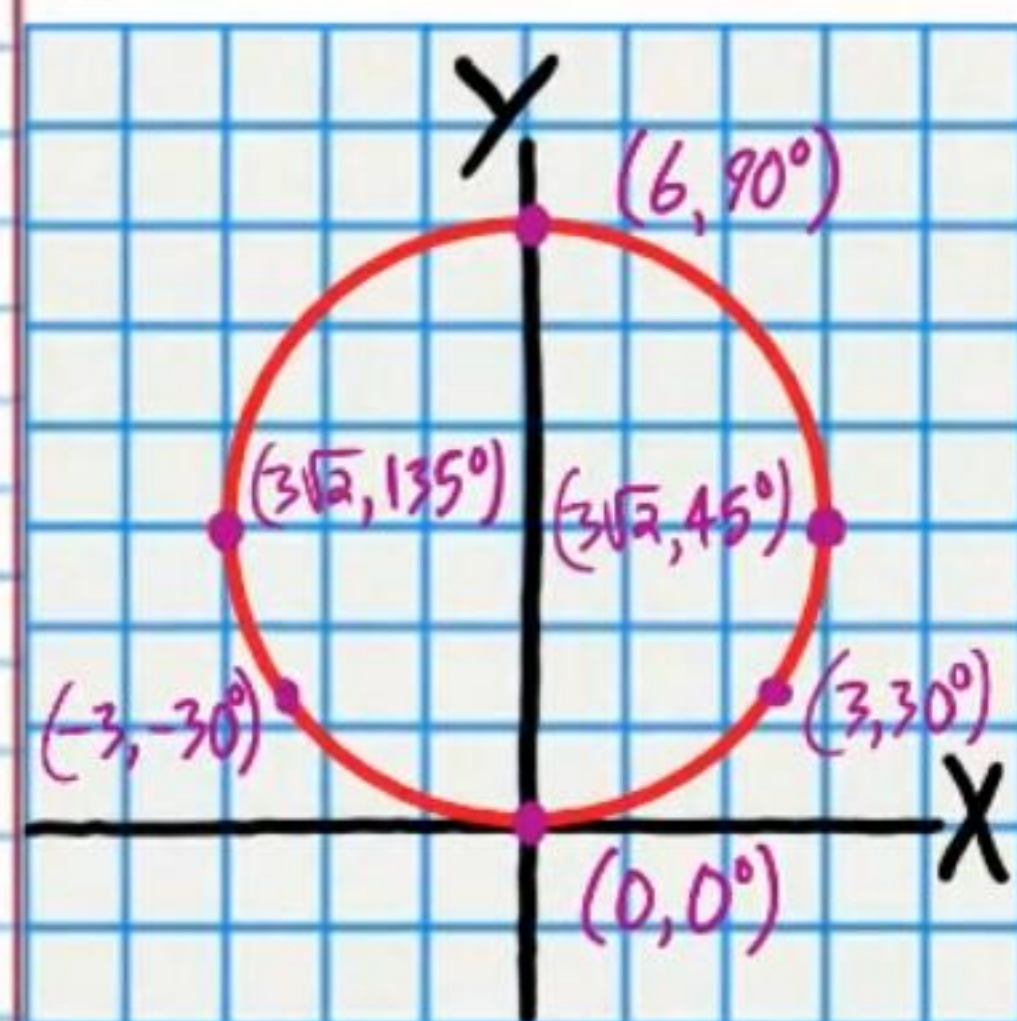
①



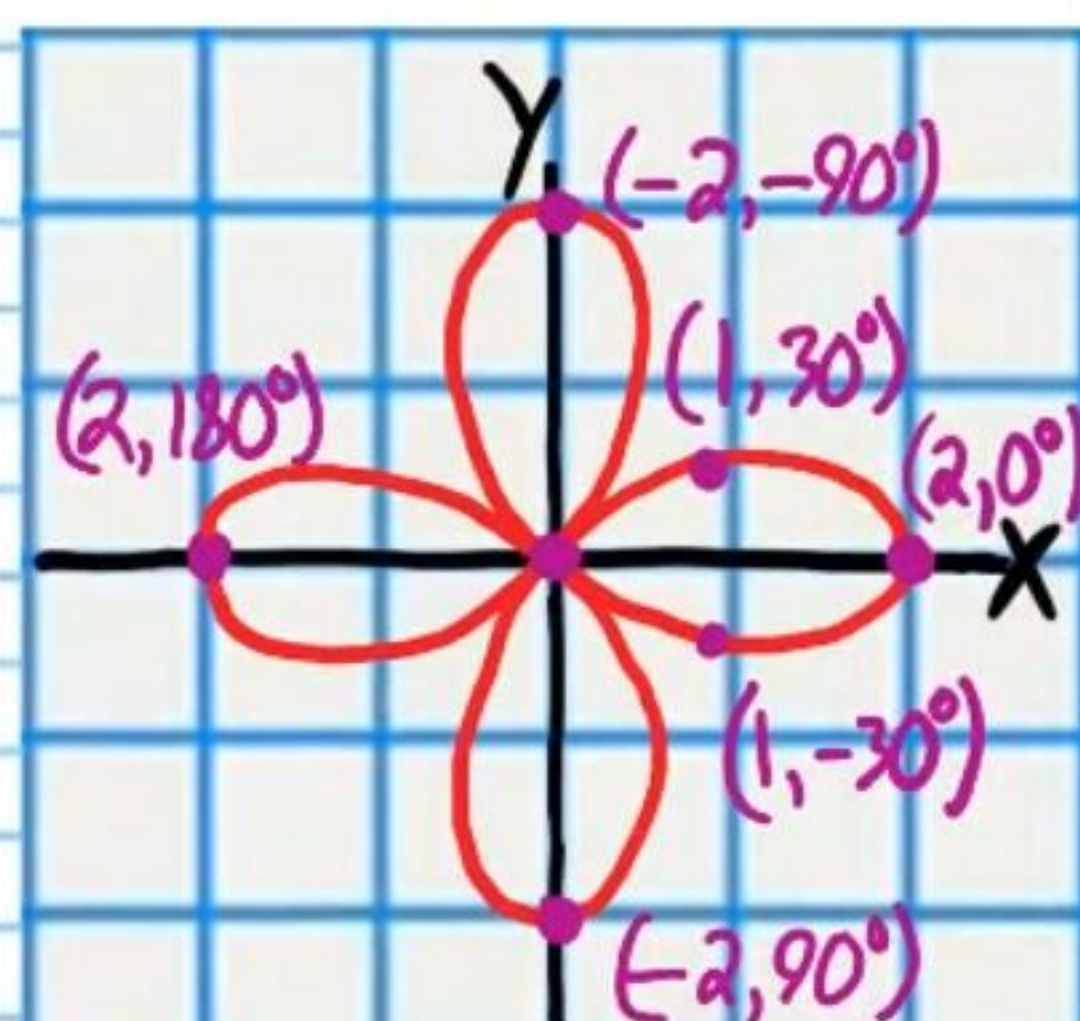
②



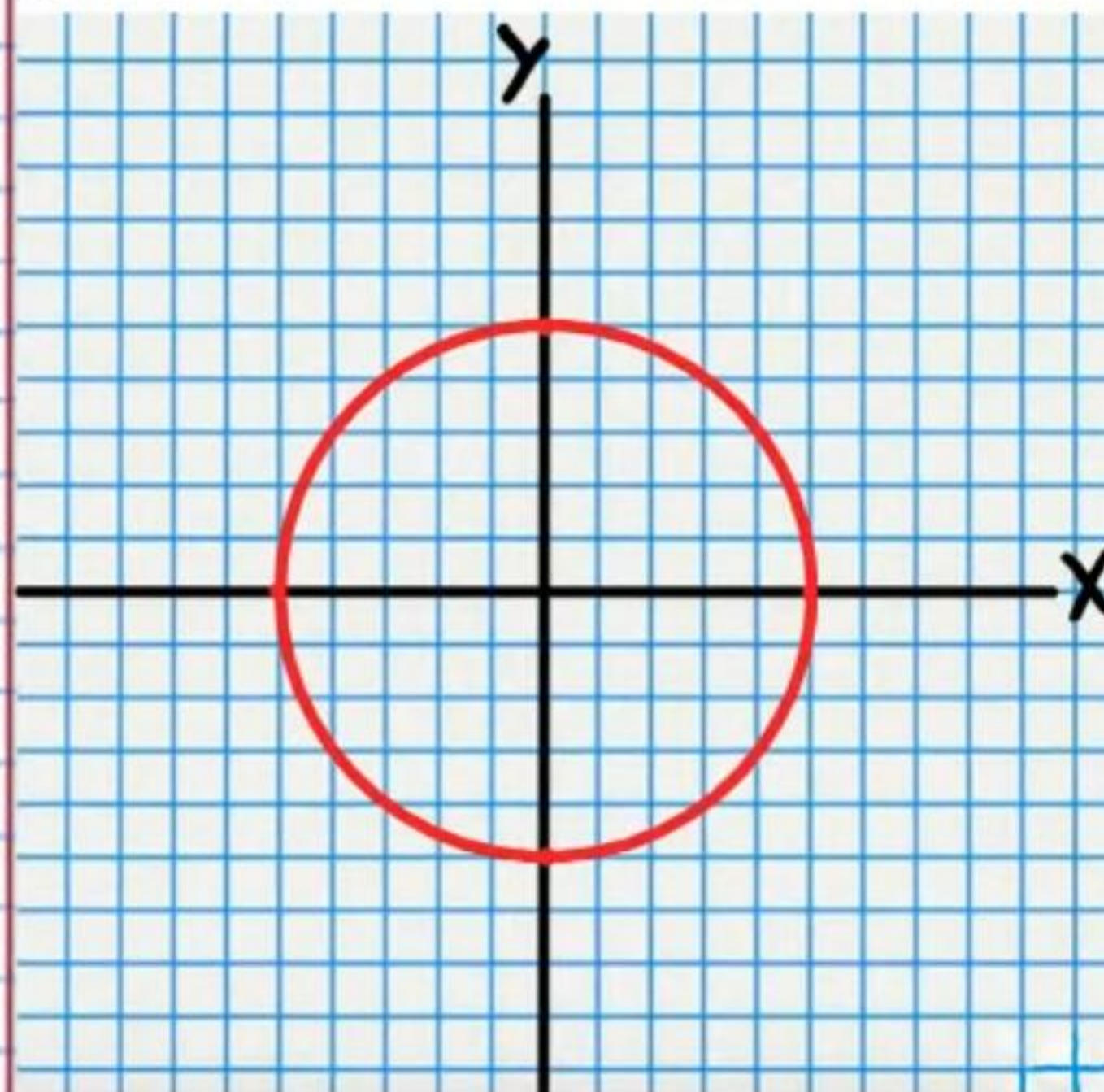
③



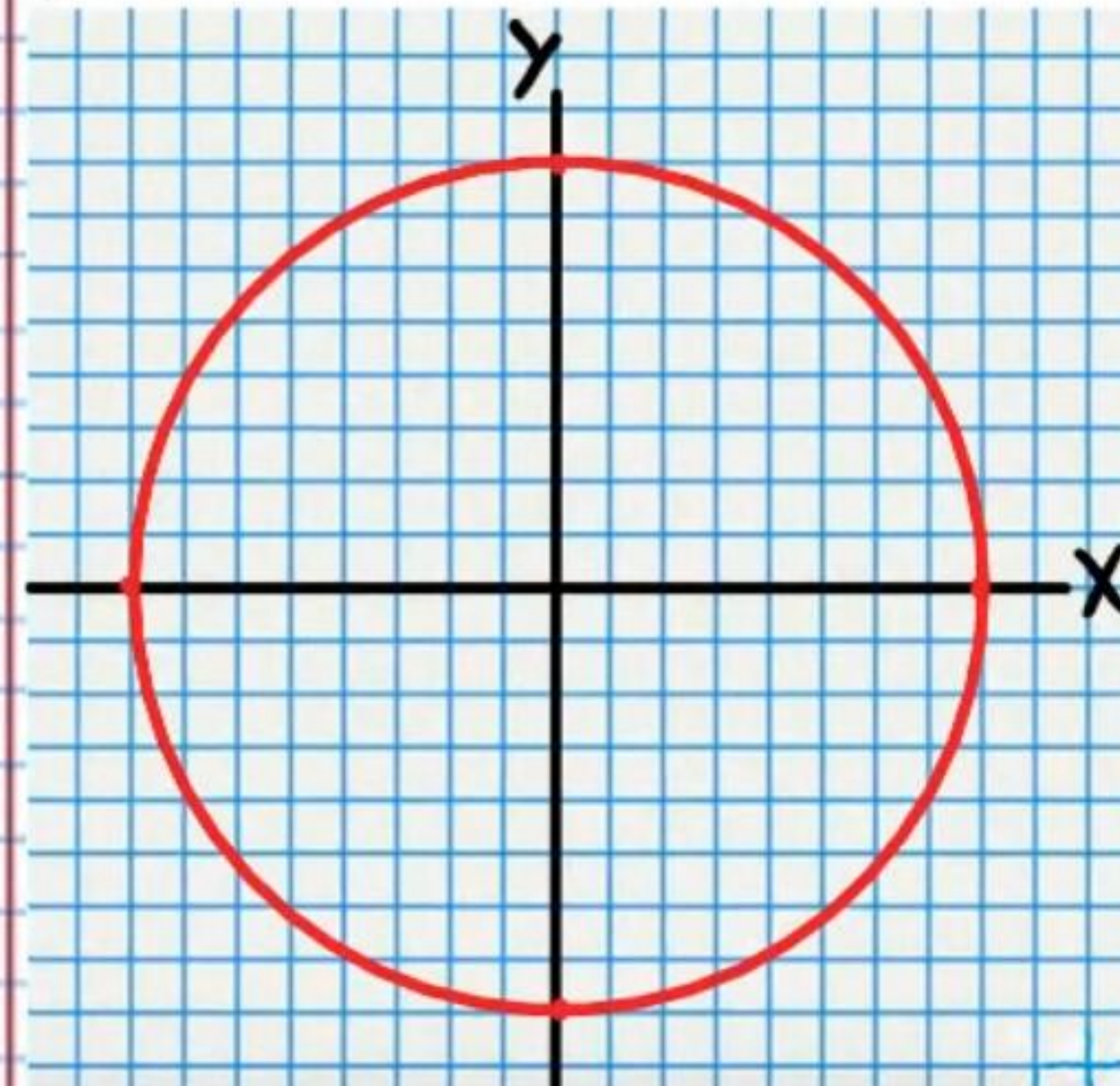
④



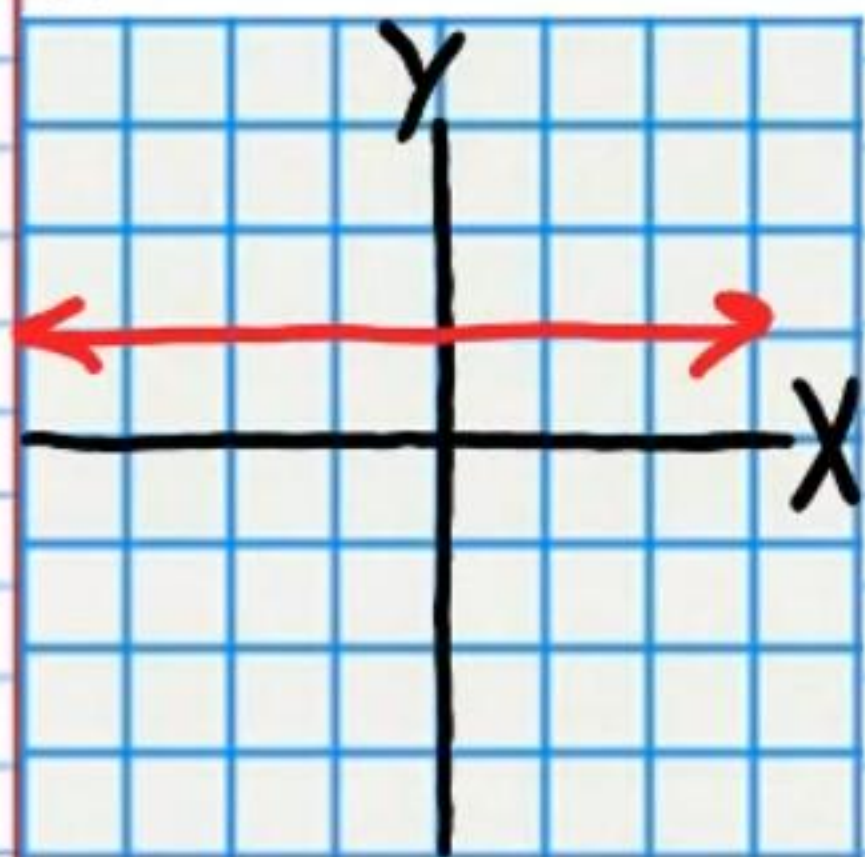
⑤



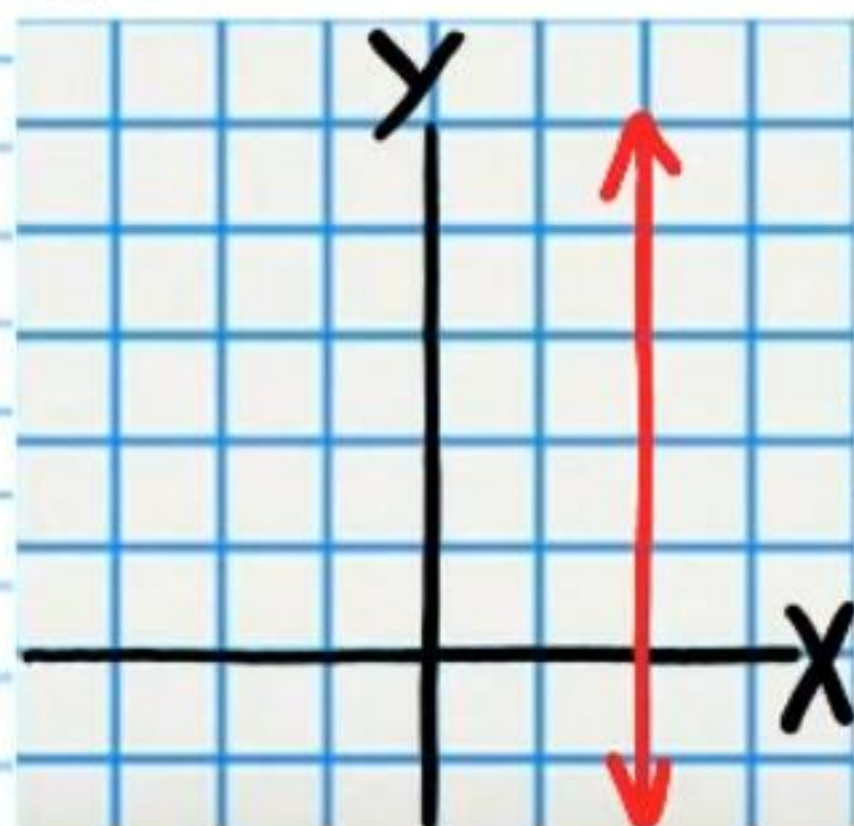
⑥



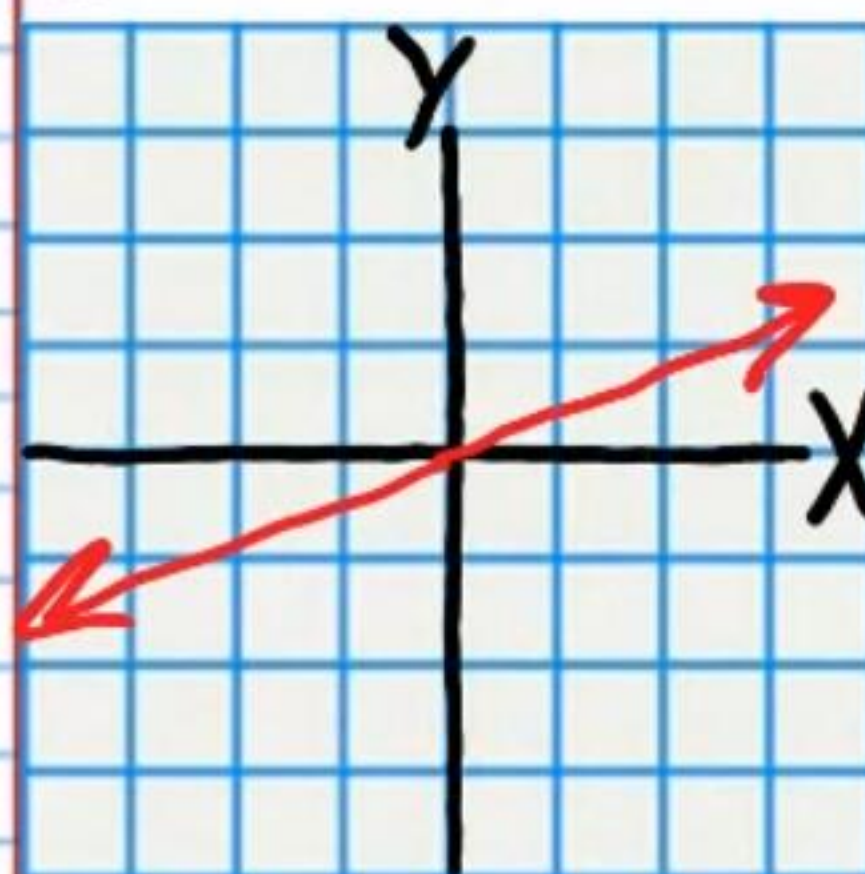
⑦



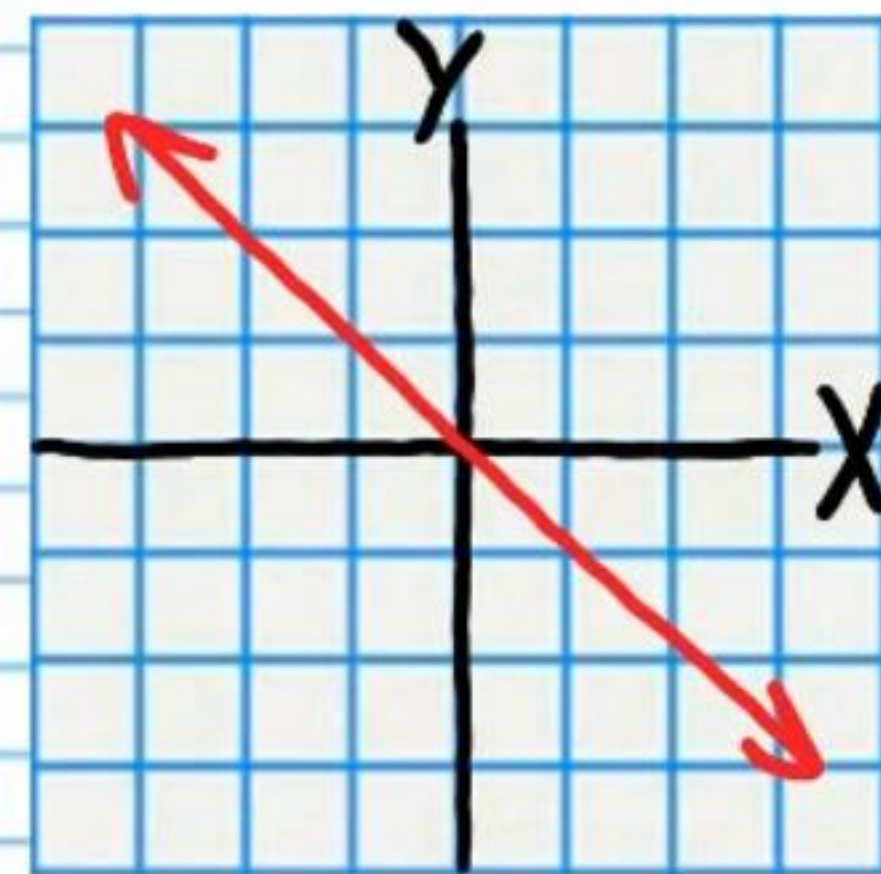
⑧



⑨



⑩



⑪ $y = \sqrt{x}$

⑬ $x + y = x^2 + y^2$

or

$$\frac{x}{\sqrt{x^2 + y^2}} + \frac{y}{\sqrt{x^2 + y^2}} = 1 + \frac{y}{x}$$

⑫ $x^2 + y^2 = 1 + \frac{y}{x}$

⑭ $x^2 + y^2 = 4\sqrt{x^2 + y^2} + y$

or

$$\left(\frac{x^2 + y^2 - y}{4} \right)^2 = x^2 + y^2$$

⑮ $r \sin \theta = 100$

or

$$r = \frac{100}{\sin \theta}$$

or

$$r = 100 \csc \theta$$

⑰ $r \sin \theta = r \cos \theta$

or

$$\sin \theta = \cos \theta$$

or

$$\frac{\sin \theta}{\cos \theta} = \frac{\cos \theta}{\cos \theta}$$

$$\tan \theta = 1$$

or

$$\frac{\sin \theta}{\sin \theta} = \frac{\cos \theta}{\sin \theta}$$

$$1 = \cot \theta$$

⑯ $4r \cos \theta + 2r \sin \theta = r^2$

or

$$4 \cos \theta + 2 \sin \theta = r$$

⑰ $1 = r^2 \sin \theta \cos \theta$

or

$$\frac{1}{\sqrt{\sin \theta \cos \theta}} = r$$

± not necessary in this context.